

Exercise 2.4

Prove that if $m^*(A) = 0$ then for each B , $m^*(A \cup B) = m^*(B)$.

Solution

We can prove equality here by establishing inequalities in both directions:

$$m^*(A \cup B) \geq m^*(B)$$

$$m^*(A \cup B) \leq m^*(B)$$

Prove that $m^*(A \cup B) \geq m^*(B)$

This direction relies on the property of monotonicity.

1. From the definition of a union, the set B is a subset of the combined set $A \cup B$:

$$B \subseteq A \cup B$$

2. Since the outer measure is monotonic we have:

$$m^*(B) \leq m^*(A \cup B)$$

Prove that $m^*(A \cup B) \leq m^*(B)$

This direction relies on the property of subadditivity.

1. The outer measure of a union is always less than or equal to the sum of the outer measures of the individual sets:

$$m^*(A \cup B) \leq m^*(A) + m^*(B)$$

2. Since we are given that $m^*(A) = 0$, we can substitute this into the inequality to get:

$$m^*(A \cup B) \leq 0 + m^*(B)$$

3. This simplifies down to the following:

$$m^*(A \cup B) \leq m^*(B)$$

We have now established the following:

1. $m^*(A \cup B) \geq m^*(B)$
2. $m^*(A \cup B) \leq m^*(B)$

Since the value has to be greater than or equal to and less than or equal to $m^*(B)$, it must be exactly equal to the following:

$$m^*(A \cup B) = m^*(B)$$

Exercise 2.5

Prove that if $m^*(A \triangle B) = 0$, then $m^*(A) = m^*(B)$.

Solution

We can start by stating the following based on the symmetric difference:

$$A \triangle B = (A \setminus B) \cup (B \setminus A)$$

We can then determine the measure of the components:

$$(A \setminus B) \subseteq (A \triangle B), \text{ then } m^*(A \setminus B) = 0$$

$$(B \setminus A) \subseteq (A \triangle B), \text{ then } m^*(B \setminus A) = 0$$

We can then prove this equality by demonstrating inequalities in both directions as follows:

$$m^*(A) \leq m^*(B)$$

$$m^*(B) \leq m^*(A)$$

Show that $m^*(A) \leq m^*(B)$

We can express set A as a subset of the union of B and the part of A that is not in B :

$$A \subseteq B \cup (A \setminus B)$$

Using the property of subadditivity of the outer measure:

$$m^*(A) \leq m^*(B) + m^*(A \setminus B)$$

If we substitute the value we found earlier, where $m^*(A \setminus B) = 0$:

$$m^*(A) \leq m^*(B) + 0$$

$$m^*(A) \leq m^*(B)$$

Show that $m^*(B) \leq m^*(A)$

We can use the same logic as above to prove this. We can express set A as a subset of the union of B and the part of A that is not in B :

$$B \subseteq A \cup (B \setminus A)$$

Using the property of subadditivity of the outer measure:

$$m^*(B) \leq m^*(A) + m^*(B \setminus A)$$

If we substitute the value we found earlier, where $m^*(B \setminus A) = 0$:

$$m^*(B) \leq m^*(A) + 0$$

$$m^*(B) \leq m^*(A)$$

Conclusion

Since we proved that $m^*(A) \leq m^*(B)$ and $m^*(B) \leq m^*(A)$, it is only logical to conclude that:

$$m^*(A) = m^*(B)$$